

Decrypt an Encrypted Message

Project description

I used Linux command-line tools to navigate directories, inspect files, and work with encrypted data. I used Linux commands to reverse a classical cipher and restore an encrypted file to its original plaintext, demonstrating practical skills in file management, command-line operations, and basic cryptography.

Read the contents of a file

I used the ls command to list the files in the current working directory and the cat command to list the contents of the README.txt file.

```
analyst@718d894ef601:~$ ls /home/analyst
Q1.encrypted README.txt  caesar
analyst@718d894ef601:~$ cat README.txt
```

Find hidden file

There was a hidden file in the caesar sub-directory, so I used the cd command to change directory to caesar sub-directory and the ls -a command to list the hidden file. I used the cat command again to list the contents of the .leftshift3 file.

```
analyst@718d894ef601:~$ cd caesar
analyst@718d894ef601:~/caesar$ ls -a
.  ..  .leftShift3
analyst@718d894ef601:~/caesar$ cat .leftShift3
Lq rughu wr uhfryhu brxu ilohv brx zloo qhhg wr hqwhu wkh iroorzlqj frppdqg:
rshqvvo dhv-256-fef -sengi2 -d -g -lq Tl.hqfubswhg -rxw Tl.uhfryhuhg -n hwwxeux
wh
analyst@718d894ef601:~/caesar$
```

To decrypt the Caesar cipher in the .leftshift3 file I used the cat command with the tr command. In this case the tr “d-za-cD-ZA-C” “a-zA-Z” translates all the lowercase

and uppercase letters in the alphabet back to their original position. The first character set, indicated by “d-za-cD-ZA-C”, is translated to the second character set, which is “a-zA-Z”.

```
analyst@718d894ef601:~/caesar$ cat .leftShift3 | tr "d-za-cD-ZA-C" "a-zA-Z"
In order to recover your files you will need to enter the following command:

openssl aes-256-cbc -pbkdf2 -a -d -in Q1.encrypted -out Q1.recovered -k ettubru
te
analyst@718d894ef601:~/caesar$
```

I then needed to return to my home directory to complete my task, so I used the cd~ command.

```
analyst@718d894ef601:~/caesar$ cd ~
analyst@718d894ef601:~$
```

Decrypt a file

I used the openssl aes-256-cbc -pbkdf2 -a -d -in Q1.encrypted -out Q1.recovered -k ettubru command to decrypt a file and recover data so I could read the message it contained.

```
analyst@718d894ef601:~$ openssl aes-256-cbc -pbkdf2 -a -d -in Q1.encrypted -out
Q1.recovered -k ettubru
analyst@718d894ef601:~$
```

In this instance, the openssl command reverses the encryption of the file with a secure symmetric cipher, as indicated by AES-256-CBC. The -pbkdf2 option is used to add extra security to the key, and -a indicates the desired encoding for the output. The -d indicates decrypting, while -in specifies the input file and -out specifies the output file. The -k specifies the password, which in this example is ettubru.

I used the ls command to list the new file Q1.recovered in the directory listing as the decrypted file that contained a message and the cat command to list the contents of the Q1.recovered file.

```
analyst@718d894ef601:~$ ls
Q1.encrypted  Q1.recovered  README.txt  caesar
analyst@718d894ef601:~$ cat Q1.recovered
If you are able to read this, then you have successfully decrypted the classic
cipher text. You recovered the encryption key that was used to encrypt this fil
e. Great work!
analyst@718d894ef601:~$
```

Summary

I learned how encryption protects information from unauthorized access and how decryption restores encrypted data to its original form. I also gained practical experience using Linux commands to navigate directories, read files, and reverse a classical cipher. This helped me understand how cryptography and command-line tools can be applied to protect and investigate data in cybersecurity.